INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

JUNEJA STREET

Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

Vihaan Juneja | Grade 9 | Cofounder | United States

Aarav Juneja | Grade 11 | Cofounder | India

github.com/V-Juneja/ml-supertrend

Investment hypothesis and complete strategy

The strategy seeks short-term rebounds after unusually strong selling in volatile stocks. A fresh low alone is not a BUY: price must confirm an upward Supertrend reversal, recent oversold RSI and signal spacing. The machine-learning layer learns online from completed hypothetical long and short probes. Three model components use rolling outcomes, adverse-movement-weighted batches and an 8 by 8 market-context memory to adjust the Supertrend multiplier and ATR smoothing within bounds. The learning changes signal sensitivity and timing; explicit portfolio rules still set entry, position size and exit. The hypothesis fails when selling reflects lasting information, or costs erase any rebound.

Each probe opens at a candle open and matures at the close of its fifth candle. A move of one entry ATR gives the long probe a score of +1 and the matching short probe -1. Only matured outcomes can update later settings. Because probes overlap, their labels are dependent and are not account returns.

At equal confidence, one ATR of adverse movement halves a probe's batch weight. Context-memory influence rises with its supporting observations but is capped at 65%. A bounded update layer smooths parameter proposals; portfolio rules separately set entries, size and exits.

Why these stocks

DELL is the paper-trading example. The strategy scans liquid S&P 500 stocks; each BUY must pass liquidity and portfolio risk checks before an order is sized.

Cofounder roles

As co-founders, Vihaan and Aarav share responsibility for strategy design, ML model review, quantitative and risk analysis, replay validation, and maintaining the research code and documentation. Both contribute across the full research cycle.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for liquid S&P 500 stocks. BUY opens one long; repeated BUYs do not add size. A confirmed SELL or protective stop closes it. Historical replay enters at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$
$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the multiplier. The boundaries sit mA above and below x; the baseline uses l = 24 and m = 1.4. They trail or reset around price. A close across the previous opposing band flips trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

The rolling learner uses up to 30 matured probes per direction: below 38% wins with mean score < 0 it proposes wider bands and longer ATR; above 62% with mean score > 0 it proposes narrower, faster settings. The batch learner weights each probe by confidence / (1 + adverse excursion in ATR units), then summarizes normalized gain, profit factor and downside. A 1-ATR adverse move halves its weight.

Context memory conditions probe outcomes on market state. Its 0–1 trend score combines 35% inverse 64-candle entropy, 35% scaled 64-candle Hurst persistence and 30% normalized ADX(14); 0 is choppy, 1 trending. Relative volatility is current ATR / trailing 100-candle mean, scaled 0–1 at 2.5×. The two axes index 8×8 = 64 cells. Each cell exponentially smooths probe gain, downside and parameter proposals (50-update half-life). A Gaussian-weighted 3×3 neighborhood forms its local proposal; sample count sets the local/global blend, capped at 65%. This adapts indicator settings, not capital allocation.

The combiner blends rolling and context proposals with available batch feedback. Larger multiplier requires a larger reversal; longer ATR responds more slowly. Deadbands, update caps, parameter bounds and baseline reversion limit drift. Separate portfolio rules control capital.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current multiplier or ATR length; theta tilde is the rounded, bounded target; alpha is the update fraction. For multiplier 1.40, target 1.50 and alpha 0.20, the update is 1.42. Deadbands ignore tiny changes; caps constrain each update and periodic reversion pulls settings toward baseline. The multiplier stays 0.5-5.0 and ATR length 5-100.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within the prior 50 candles and at least 10 candles since the last signal. RSI compares Wilder-smoothed gains and losses over 14 candles on a 0–100 scale. SELL requires a fresh-high reversal, downward flip and recent RSI above 70. Signals use completed candles; volume and tick-pressure filters are off.

Implementation check. Yahoo data: 780 DELL bars, 776 matured probes per side; multiplier 1.40 to 2.145, ATR stayed 24; 6 BUY / 8 SELL. Prefix outputs matched.

2 ML Supertrend replay and backtest charts

We replayed ML Supertrend BUY emails from 8 September 2025 to 14 October 2026, carrying the last list through 30 October (290 sessions). Each list was applied at the next session open, starting with $100,000 and charging 10 basis points per side. Historical sizing allowed up to ten holdings; the current declaration caps the portfolio at five.

ML Supertrend replay result	Value
Ending marked equity	$151,542.68
Total return / annualized growth	+51.54% / +43.92%
Maximum drawdown	-21.10%
QQQ benchmark return	+19.39%

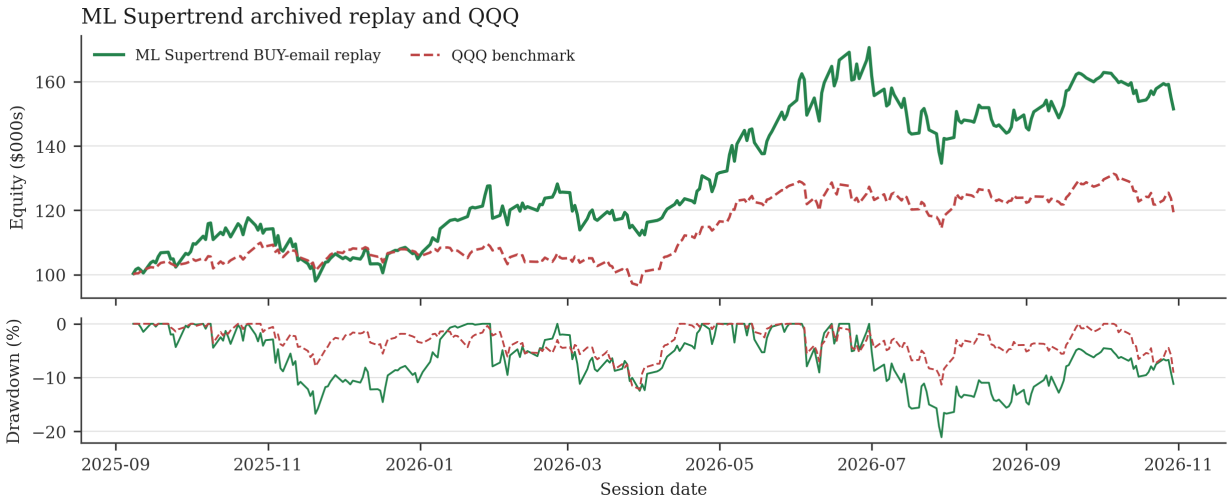


Figure 1. Ending equity was $151.5k versus QQQ at $119.4k; the 21.10% maximum drawdown shows returns came with sharp setbacks. Data: Alpaca SIP split-adjusted bars; our replay.

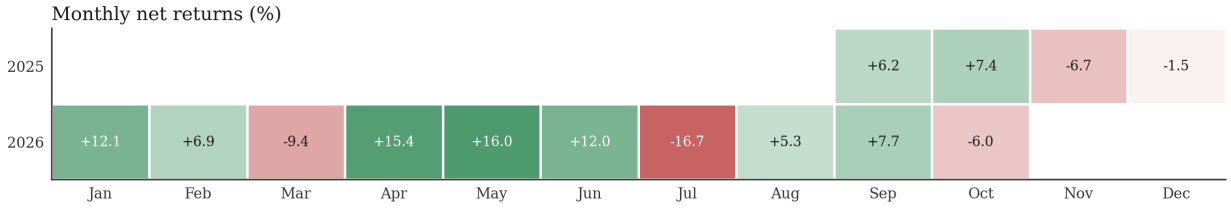


Figure 2. Nine of 14 months gained; July 2026 lost 16.7%, showing uneven performance. September 2025 is partial. Data: same replay.

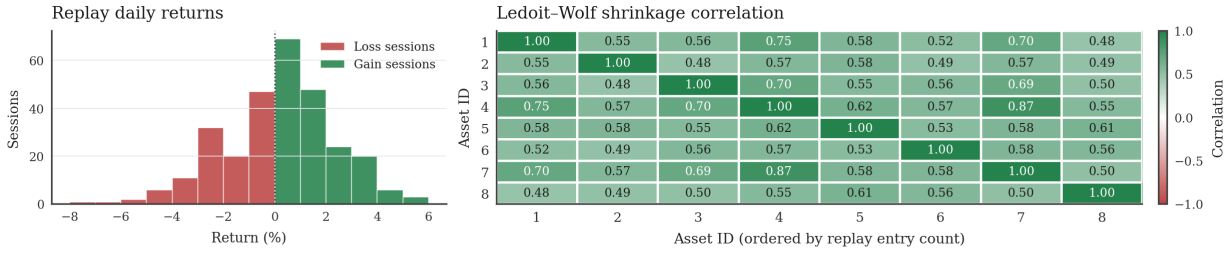


Figure 3. Red bars are loss sessions. IDs 1–8 are actual stocks ranked by archived BUY-entry count (1 most frequent), not random; ticker names are omitted. Correlations +0.48 to +0.87 show the names co-moved. Data: same 289 replay returns.

Sensitivity: +39.11% at 20 bps; +43.49% with one-session delay. All 1,254 fills and 290 marks reproduce. Method · GitHub.

3 Trade expectancy and risk tests

Net Trade Expectancy Equation

$$e = pW - (1-p)D, \qquad p_{BE} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

Trade statistic	Historical result
Completed trades	622
Win rate	46.62%
Mean net profit / return per trade	+$81.73 / +0.8173%
Median trade return	-0.6268%
Average net winning / losing return	+8.05% / -5.50%
Profit factor	1.28
Break-even win rate at observed payoffs	40.59%
Best / worst net trade return	+62.80% / -40.03%
Longest losing streak	15 trades

These 622 closed trades come from the ML Supertrend email replay. The 46.62% win rate exceeded its 40.59% break-even level; profit factor was 1.28. The median trade lost 0.6268%, while larger gains lifted the mean. Open positions are excluded. Trade calculations and fills: GitHub trade evidence.

ML Supertrend one-year Monte Carlo

We resampled the daily equity returns from the archived ML Supertrend replay: 290 sessions, 8 September 2025–30 October 2026. Circular five-session blocks preserve short runs of gains and losses. The 50,000 paths each span 252 sessions. The replay source, calculations and chart are documented in the Monte Carlo files.

One-year path statistic	Result
Median terminal return	+43.98%
5th / 95th percentile terminal return	-11.85% / +131.43%
Paths ending below start	10.87%
Mean return in worst 5% of paths	-21.93%
95th percentile maximum drawdown	35.46%

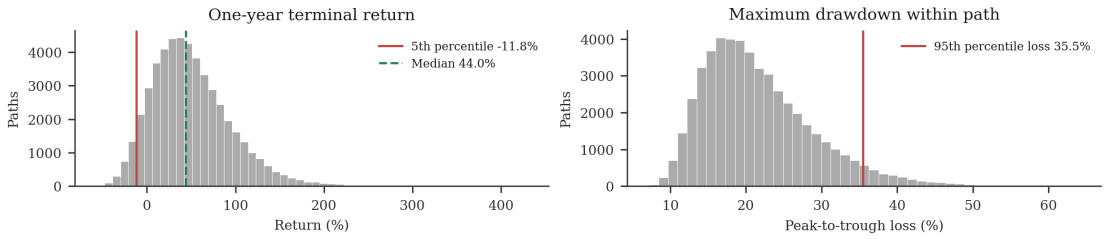


Figure 4. Across resampled 252-session paths, median return was +43.98%, but 10.87% ended below the starting value; the 5th-percentile return was -11.85% and 95th-percentile maximum drawdown was 35.46%. This spread shows substantial downside despite the positive median.

How to read Figure 3: Ledoit–Wolf reduces noise by blending observed covariance (how returns move together) with a reference that assumes equal variance and zero pairwise covariance. Rescaling by both volatilities gives correlation: -1 means opposite moves, +1 the same direction. The +0.48 to +0.87 results show the selected stocks moved together, limiting diversification. This is descriptive, not a signal. Method and chart data.

4 Portfolio construction and risk

False reversals can cluster, overnight gaps can cross stops and technology names can fall together. Entry spacing, stop-risk budgets, concentration caps and account-loss rules address these strategy-specific risks. Limits are declared choices, not optimized. Risk-test assumptions and code.

Control	Execution rule and purpose
Exposure	Long-only, no borrowing; at most five names, 15% of equity per name and 75% gross stocks. Five names at the individual cap use the full portfolio allowance.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Execution integrity	Use one account; reject duplicate BUY entries. Log accepted/rejected signals, settings, quotes, order and fill times, fees, stop and exit reason. Freeze the declared rules; close by the competition cutoff.

Position Sizing Equation

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.15E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At the recorded DELL close of $563.17, a $15,000 cap permits 26 shares ($14,642.42). The $500 loss budget in a $100,000 account allows at most $19.23 stop distance at that size. A wider two-ATR stop reduces shares. The 15% name limit and five-position limit allow at most 75% gross exposure; remaining equity stays in cash. Position size can be lower when the stop-risk budget binds.

Gap and execution stress

Stop execution stress	Loss / planned stop loss	Account loss at 2% risk budget
Fills at intended stop	1.0 x	2%
Loss exceeds stop distance by 50%	1.5 x	3%
Loss doubles the intended distance	2.0 x	4%
Gap produces three times planned loss	3.0 x	6%

The table multiplies planned account risk by the realized stop-distance overrun. A gap can exceed every scenario shown, so the 6% halt trigger cannot cap losses.

Stress grid: 96 combinations of horizon, exposure, volatility and correlation (100,000 paths each); daily Gaussian, t(5), gap and liquidation tests use 40,000 paths per model. At the 75% ceiling, common 10%/20% price declines produce 7.5%/15% account losses before costs. Opening gaps can exceed the 6% halt threshold.

Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold at most five names, 15% per name and 75% total stocks. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.